Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	KALIDERES
Project / Solution Name	FlowUp
Name (Leader)	CARLOS LEE
Email & Phone (Leader)	Carlosleetest2@gmail.com +60 176484186
University Affiliation(s)	Asia Pacific University
Live Prototype URL	https://flowup-mvp.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/lee-carlos/SDCHACK_KALIDERES_1

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Flow: user → web app → backend → database → AI → human approval, hosted on the cloud.
Users and app: owner, cashier, supply, finance and analyst use the FlowUp web app (React + TypeScript), published through Replit Deployments.
Backend: a Python FastAPI service runs rule-based logic that calculates every number (stock reminder, EOQ, holding cost, supplier score) and runs the reconciliation check: opening stock + goods in − cost of goods sold = closing stock.
Database: synthetic store data in a Replit-hosted PostgreSQL database.
AI: limited, structured results go to the AI integration built into Replit. There is no separate AI service. It returns short explanations and supplier message drafts, with a template fallback if it is unavailable.
Human approval: a person verifies or corrects flagged data, and every action is kept in an audit log.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
PostgreSQL	Relational database for products, sales, stock, suppliers and finance	PostgreSQL License
React	Web interface, dashboards and charts	MIT
TypeScript	Typed language for the frontend	Apache-2.0
Python + FastAPI	Backend API and rule-based calculation logic	PSF License / MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Replit AI integration (Replit)	Writes short explanations and supplier message drafts from rule results	Sponsor-provided credits plus a self-paid plan

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Deployments (Replit)	Hosts the live prototype (currently private, access arranged for reviewers)	Sponsor credits plus a self-paid plan
Replit-hosted PostgreSQL (Replit)	Stores synthetic store data	Sponsor credits plus a self-paid plan

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Model provided through the Replit AI integration	Replit	Pre-trained, used as provided	Replit and model provider terms
None	No trained model is used for numbers. Stock reminders, EOQ, holding cost and supplier score use rules.	N/A	N/A

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic store data	Created by the team, stored in the Replit PostgreSQL database	Synthetic	Team-owned	N
Reference test dataset	Created by the team, used to check formulas and reconciliation	Synthetic	Team-owned	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit Agent	Generated the app from our structured specification, then refined in the Replit workspace	Major
Claude	Writing and rewriting submission documents	Minor
ChatGPT	Drafting and ideas	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

We wrote a structured specification (roles, data flow, formulas, reference test numbers) before prompting Replit Agent, then reviewed and refined the generated app in the Replit workspace. Every formula is defined once and checked against a reference dataset and a reconciliation check. Rules decide the numbers, AI explains it. Then, human approval and an audit log keep a person responsible for every action.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Credits for Replit Agent, Deployments, hosted PostgreSQL and the AI integration	Built the app, hosts the live prototype, stores the synthetic data and powers the AI explanations

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is essential to how the prototype is built and hosted today. The core features (rule-based calculations, cashier flow, stock updates, audit log) do not depend on the AI part. The AI part only writes explanations and message drafts, with a template fallback, so it can be replaced without breaking the core.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The prototype runs on synthetic store data only. There is no real customer data is collected, so there is nothing real to retain. Only limited, structured rule results (for example stock figures and supplier scores) are sent to the AI integration, never unrestricted database access. Access is role-based, and every action is kept in an audit log. The deployment is currently private.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Hallucination: AI never calculates. Numbers come from rules, and AI only writes short explanations and supplier drafts, labelled as suggestions.
Availability: if the AI is unavailable, a template fallback is used.
Single provider: AI, hosting and database all depend on Replit.
Synthetic data: test results are prototype results, not proof of real-store gains.
Scope: the MVP covers one journey. Other modules are on the roadmap.
Private deployment: access must be arranged for judges.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☐ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



Carlos Lee

25/10/2026